



Optimal eVTOL Charging and Passenger Serving Scheduling for On-Demand Urban Air Mobility

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One of the critical obstacles that restricts the large-scale urban air mobility (UAM) service is limited flight range of electric vertical takeoff and landing aircrafts (eVTOL). Continuous operations as well as on-demand service require frequent recharge of eVTOLs within a limited time. On the one hand, it puts high requirements on battery energy storage and facility charging rate. On the other hand, a well-designed scheduling strategy should be implemented to satisfy the recharging needs of eVTOL while satisfying the service demand from UAM users. In our research, a mixed integer program is developed to model the eVTOL charging scheduling process with the objective to minimize the total waiting time of passengers. The proposed model is able to solve the charging and passenger serving scheduling problem in a reasonable short time, making it feasible for real-time operations of UAM service. Meanwhile, the result analyses provide implications regarding the relations among the number of charging facilities, computation efficiency and the service quality.

I. Nomenclature

J	= Set of eVTOL aircrafts, indexed $j = 1, \dots, n$
μ	= Set of charging facilities, indexed $i = 1, \dots, m$
τ	= Set of time periods, indexed $t = 1, \dots, l$, time period t starts from time $t - 1$ and ends at time t
K	= Set of passengers, index $k = 1, \dots, w$
SoC_j	= State of charge for eVTOL j upon arrival at vertiport
SoC_{min}	= Minimum state of charge for eVTOL j before takeoff
LoS	= Predefined level of service, i.e. the maximum waiting time for each passenger
c	= Capacity of eVTOL battery
q	= eVTOL charging rate
p_j	= Charging time for eVTOL j
r_j	= Release time for eVTOL j to get charged (equals arrival time of eVTOL j)
d_k	= Arrival time of passenger k at vertiport
w_{kq}	= Indicator for the case if passengers k, q share the same destination
l	= Upper bound for makespan (total time required to charge all eVTOL)
M	= A large number (i.e., "big-M")
C_j	= Charging completion time for eVTOL j
F_j	= eVTOL takeoff time
T_{jk}	= Tardiness for passenger k taking eVTOL j , i.e. delay experienced by passenger k
x_{ij}^t	= Binary decision variable and takes value of 1 if eVTOL j get charged at facility i starting from period t , 0 otherwise.
y_{jk}	= Binary decision variable and takes value of 1 if eVTOL j is to serve customer k , 0 otherwise.

II. Introduction

Urban Air Mobility (UAM) is an emerging concept in recent decade and has been widely discussed from policy-making authorities, industry companies, manufactures to academic researchers. UAM is defined as safe and efficient air traffic operations in metropolitan areas for manned and unmanned aircraft systems with objective to mitigate severe traffic congestion issues and greatly improve urban mobility [1]. The concept of UAM can date back to the 1960s,

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when several companies used aircraft to provide point-to-point commuting service within and around metropolitan areas in Los Angeles, San Francisco, New York and Chicago [2]. Its recurrence contributes to the technology breakthrough on the development of eVTOL aircrafts, which integrates advanced autonomous and distributed electric propulsion technologies and is expected to provide safer, quieter, and more efficient air transportation service in urban areas. Except for vast investment in vehicle design and propulsion technologies from numerous manufactures [3-5], efforts are also devoted to studies on UAM concept definition, feasibility study, network design, and operation scheduling [6-11].

In this paper, we develop a mixed integer programming model to represent eVTOL charging as well as passenger serving scheduling process at vertiports (i.e. hubs where eVTOL takeoff, land, pick up passengers, and get charged). The proposed model is able to obtain the optimal scheduling solution within a reasonable short time under different scheduling horizon scenarios. This is critical as we are laying foundations for real-time operations of eVTOL aircrafts such that large-scale on-demand UAM service will be feasible. Our proposed model is validated by different instances and scenarios. Insights regarding relations between charging facility capacity, computation efficiency and service quality are also discussed.

III. Motivation

One of the key characteristics of UAM service is that the electric propulsion is the main propulsion choice for VTOL aircraft. However, the specific energy (the amount of energy per unit weight provided by the battery) of Lithium-ion polymer (LiPo) batteries today is insufficient for long range commutes [12] and they have to frequently get charged under rapid charging capability to ensure continuous operations. Uber has proposed that the eVTOL must be able to operate continuously for at least 3 hours while flying 25 mile missions at the battery's end-of-life state to ensure large-scale UAM operations, which may need charging capacity of up to 600 kW and require vehicle get charged continuously between two flights [13]. Fig. 1 illustrates expected eVTOL charge and discharge process to satisfy large-scale on-demand UAM service [14].

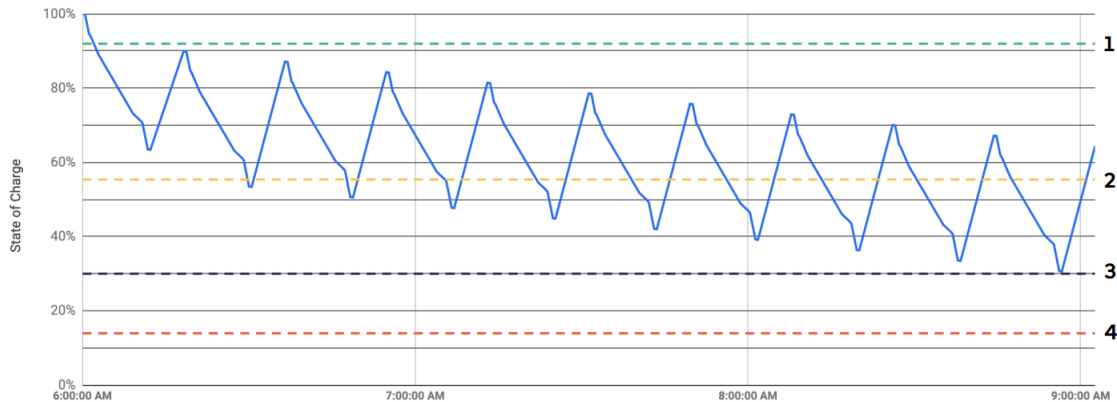


Fig. 1 EVTOL charging and discharge process within a three hour period

Notes:

1. Battery Rapid Charge Ceiling: Above an upper state of charge (SoC), the charge rate will begin to decrease from maximum capability.
2. Minimum State of Charge for Dispatch: Battery must be above this SoC to takeoff.
3. Minimum Reserve State of Charge: Battery SoC should not pass below this line in nominal operations.
4. At low SoC conditions, the current spikes and voltage drops precipitously.

Considering the energy sensitive operations of eVTOL aircrafts, several studies have worked on identification of optimal arrival trajectories and schedule of eVTOL arrival sequence with consideration of aircrafts' state of charge (SoC) [7, 12]. Nevertheless, no significant research has studied the sequencing and scheduling of eVTOL charging at vertiports with constraints to satisfy the users' request for on-demand service, required SoC operation level and limited number of charging facilities. This paper aims to tackle this problem by design the charging schedules to minimize the delay of passengers while satisfy the battery-charging needs of eVTOL.

IV. EVTOL Charging Sequencing and Scheduling Modeling

A. Assumptions

Before discussion of the modeling approach for eVTOL charging scheduling, several fundamental assumptions are described as follows:

1. Charging Facilities are all identical. They provide the same charging rate for each eVTOL and each is able to charge only one eVTOL at one time period.
2. All eVTOLs are identical, namely with same battery capacity and charging efficiency and each of them must get charged to a predefined SoC level before serving passengers.
3. There is enough capacity at vertiports for arriving eVTOL to wait to get charged.
4. Each service request from passenger must be satisfied and the waiting time for each passenger should not beyond a predefined threshold to ensure service quality.
5. Arrival time of each passenger and eVTOL within the schedule horizon is known in advance as well as the SoC of eVTOLs upon arrival at vertiport.
6. Each eVTOL is ready to get charged once upon arrival and is able to serve at most four passengers at one time.

B. Modeling Approach

Figure 2 describes the simplified UAM operation process at a vertiport. The vertiport is equipped with several charging facilities. Upon arrival at vertiport at time r_j , each eVTOL j must get charged if SoC_j is below the predefined level SoC_{min} before serving passengers. The charging time can be calculated as follows:

$$p_j = c * (SoC_{min} - SoC_j) / q \quad (1)$$

If SoC_j is larger than SoC_{min} , the eVTOL do not need to get charged, p_j will be zero. If eVTOL starts to get charged from time t , then the time eVTOL j is ready to serve passengers will $C_j, C_j = t + p_j$.

After finishing charging, each eVTOL needs to serve up to four passengers. There are two scenarios that can occur. In the first scenario, C_j is earlier than the arrival time of last passenger k that is assigned to be served by eVTOL j . Then the eVTOL takeoff time will be the arrival time of this passenger k, d_k . For the second scenario, C_j is larger than the arrival time of any passenger that will be served by eVTOL j . In this case, eVTOL takeoff time will be the charging completion time C_j . The tardiness or waiting time for each passenger can be obtained by the difference of eVTOL aircraft takeoff time and passenger arrival time. It should be mentioned that we do not consider the operation time of other operations such as passenger on-boarding, disembark, and eVTOL taxi-in and taxi-out. These operation time can be considered as a constant added to the proposed formulation and will not influence the final schedule.

Meanwhile, eVTOLs that arrive at the studied vertiport may be divided into two types. Type A eVTOL are those serving passengers with studied vertiport as destination. However, those eVTOLs may not be enough to satisfy all the demand at the studied vertiport. Therefore, we may dispatch available eVTOL from other vertiports to fulfill the service, which is Type B eVTOL as presented in Fig. 2.

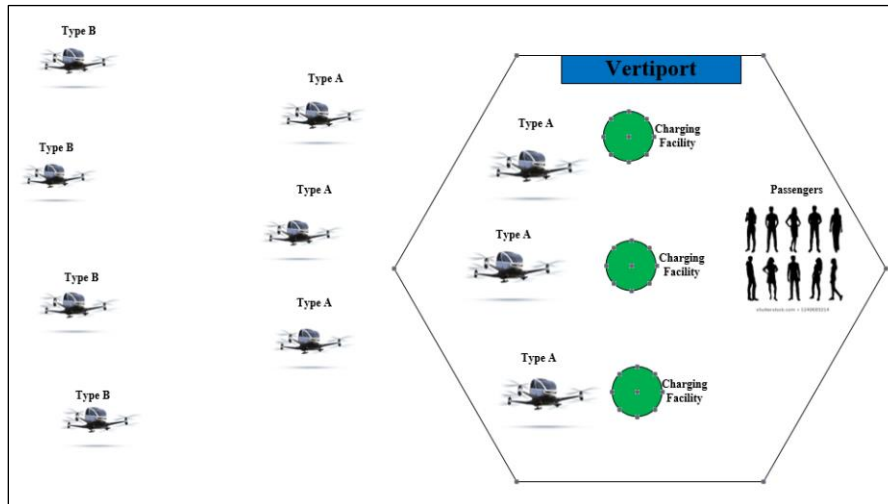


Fig. 2 EVTOL charging and passenger serving scheduling process

The modeling structure of the scheduling problem in this study can be decomposed into two sub-models, eVTOL charging scheduling model and passenger serving scheduling model. First of all, we discretize the studied time horizon into a set of time periods. For the eVTOL charging scheduling model, one eVTOL can start to get charged at one of those time periods at one charging station, which can be described by constraint (2). In addition, for each charging facility, at any period of time, only one eVTOL can be in charging process as represented by constraint (3). Besides, eVTOL can only start to get charged after release time, which is restricted by constraint (4).

$$\sum_{i \in \mu} \sum_{t=1}^{l-p_j+1} x_{ij}^t = 1, j \in J \quad (2)$$

$$\sum_{j \in J} \sum_{s=\max(1, t-p_j+1)}^t x_{ij}^s \leq 1, i \in \mu, t \in \tau \quad (3)$$

$$\sum_{i \in \mu} \sum_{t=1}^{r_j} x_{ij}^t = 0, j \in J \quad (4)$$

For passenger scheduling model, each passenger can only be served by exactly one eVTOL while each eVTOL can take at most four passengers, which are represented by constraints (5) and (6). It should be noted that passengers can be served by the same eVTOL only when they share the same destination vertiport, which is restricted by constraint (7).

$$\sum_{j \in J} y_{jk} = 1, k \in K \quad (5)$$

$$\sum_{k \in K} y_{jk} \leq 4, j \in J \quad (6)$$

$$y_{jk} + y_{jq} - 1 \leq w_{kq}, j \in J, q, k \in K, q \neq k \quad (7)$$

To compute tardiness for each of the passengers, we need first to obtain the takeoff time of eVTOL serving that passenger. The takeoff time should be greater than the charging completion time of eVTOL and greater than the arrival time of any passenger served by that eVTOL, which are expressed as constraint (8) and (9) respectively. The completion time of any eVTOL can be represented by inequality (10) and therefore the tardiness of each passenger can be represented by inequality (11) and restricted by predefined level of service.

$$F_j \geq C_j, j \in J \quad (8)$$

$$F_j \geq d_k y_{jk}, j \in J, k \in K \quad (9)$$

$$C_j \geq \sum_{i \in \mu} \sum_{t=1}^{l-p_j+1} (t + p_j - 1) x_{ij}^t, j \in J \quad (10)$$

$$T_{jk} \geq F_j - d_k - M(1 - y_{jk}), j \in J, k \in K \quad (11)$$

$$T_{jk} \leq LoS, j \in J, k \in K \quad (12)$$

Our objective function is to minimize the total tardiness or total waiting time for all passengers, which can be expressed as:

$$\text{Min} \sum_k \sum_j T_{jk}$$

Based on all the discussion above, we developed a scheduling optimization model for eVTOL charging and serving passengers. For completeness, we repeat some previously presented constraints here:

$$\text{Min} \sum_k \sum_j T_{jk}$$

s. t.

$$\sum_{i \in \mu} \sum_{t=1}^{l-p_j+1} x_{ij}^t = 1, j \in J$$

$$\sum_{j \in J} \sum_{s=\max(1, t-p_j+1)}^t x_{ij}^s \leq 1, i \in \mu, t \in \tau$$

$$\sum_{i \in \mu} \sum_{t=1}^{r_j} x_{ij}^t = 0, j \in J$$

$$\sum_{j \in J} y_{jk} = 1, k \in K$$

$$\begin{aligned}
\sum_{k \in K} y_{jk} &\leq 4, j \in J \\
y_{jk} + y_{jq} - 1 &\leq w_{kq}, j \in J, q, k \in K, q \neq k \\
F_j &\geq C_j, j \in J \\
F_j &\geq d_k y_{jk}, j \in J, k \in K \\
C_j &\geq \sum_{i \in \mu} \sum_{t=1}^{l-p_j+1} (t + p_j - 1) x_{ij}^t, j \in J \\
T_{jk} &\geq F_j - d_k - M(1 - y_{jk}), j \in J, k \in K \\
T_{jk} &\leq LoS, j \in j, k \in K \\
x_{ij}^t &\in \{0, 1\}, y_{jk} \in \{0, 1\}, C_j \geq 0, T_{jk} \geq 0, i \in \mu, t \in \tau, j \in J, k \in K
\end{aligned}$$

V. Numerical Study

From authors' another research, we have developed a mathematical model to identify optimal locations for vertiports. Based on a case study in Tampa Bay area in Florida, U.S., we are able to estimate the potential demand of UAM users and allocate optimal travel route for each user. In this research, we are going to use the busiest vertiport identified in that research for the numerical study. In total, there are 508 eVTOL arrivals with the studied vertiport as the destination and 505 passenger arrivals to start their trips from this vertiport for a typical weekday. Their corresponding distributions are presented in Fig 3 and Fig 4. These distribution characteristics are consistent with what is simulated in other literatures[15]. For the battery capacity, Uber claimed the expected value to be at least 140 kWh and appeared to be higher in their later reported vehicle requirement for eVTOL [14, 16]. In this paper, we take value of 150 kWh. In addition, we take 0.7 as the minimum SoC for eVTOL, which can approximately support a 60 mile mission considering the minimum reserve SoC and eVTOL takeoff and landing energy consumption. As for SoC for each arriving eVTOL, they are generated randomly.

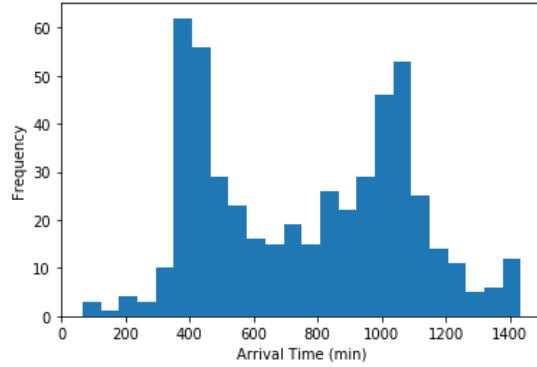


Fig. 3 Passenger arrival time distribution for 24 hours

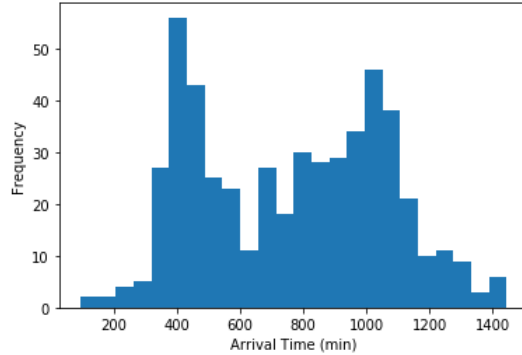


Fig. 4 eVTOL arrival time distribution for 24 hours



In the base case scenario, we consider there are 3 charging stations in the studied vertiport. Different instances are firstly tested for different scheduling horizons from 5 to 30 minutes. As assumed above, we can obtain all the eVTOL and passenger arrival information within the schedule horizons. The studied period will start from 7:00 am (i.e. 420 min in the above figure), which is within the morning peak hours. To ensure that each customer can be served, if the number of eVTOL arrivals (Type A) are less than the number of passengers, we assume that there is some backup eVTOL (Type B), the number of which equals the difference between the number of passengers and the number of Type A eVTOL. The optimization problems are solved by Gurobi solver and coded in Python programming language in a Dell computer with 64-bit operating system, 16.0 GB RAM and Intel Core i7-7700 CPU. Results are presented in Table 1.

Under current setting, it can be observed that the solution time is within 3 minutes for all scheduling horizons. When scheduling horizon is within 10 minutes, the solution time is quite fast (within 20 seconds). With scheduling horizon increases, the computation time does not vary significantly and is close to 3 minutes. However, we can also notice that the total delay for passengers does not have increasing trend as well while the number of passenger arrivals keeps growing, which indicate that the average delays for each passenger is decreasing with increasing schedule horizon. This is understandable because we are able to collect more information for both eVTOLs and passengers with increasing schedule horizon, which provides us more flexibility and options to find better schedules. Besides, 3 minutes is also an acceptable real-time schedule time considering a 30 minute schedule horizon. Therefore, longer schedule horizon is suggested in operation from this perspective.

Table 1 Optimization results under different scheduling horizons

Schedule horizon (min)	Number of Charging Facility	Number of passenger arrivals	Number of eVTOL arrivals	Number of backup eVTOL	Computation time (s)	Optimal objective value (min)
5	3	14	7	7	10	52
10	3	18	14	4	15	47
15	3	21	16	5	175	54
20	3	27	17	10	163	46
25	3	28	19	9	174	48
30	3	33	20	13	160	44

Besides, we are also interested in the effect of increasing number of charging facilities on our solution results. Therefore, under the base scenario of 30 minutes' scheduling horizon and 3 charging facilities, several instance test results are illustrated in Table 2. From Table 2, we can see that on the one hand, when the number of charging facilities are more than 4, the computation time is quite fast (less than 20 seconds) with no total delay increase. This result reveals that enough number of charging facility can help in computation efficiency of the proposed model as it is easier to find the optimal schedule. On the other hand, increasing the number of charging facilities does not necessarily have positive effect on improving service quality. In this case, the delays for passengers may result from lack of supplies of eVTOL. In addition, we also test the cases when there are only one or two charging facilities. With the limited running time of 6 minutes, scenario with two charging facilities is only able to find solution with optimality gap of 38.46% and objective value of 65 minutes. For scenario with only one charging facility, it is even hard to find an upper bound of total delays within 6 minutes. This reveals that lack of charging facilities will severely damage the efficiency of real-time operation scheduling as it will be hard to find an optimal solution within a short time.

Table 2 Optimization results with different number of charging facilities

Schedule horizon (min)	Number of Charging Facility	Number of passenger arrivals	Number of eVTOL arrivals	Number of backup eVTOL	Computation time (s)	Objective value (min)
30	3	33	20	13	160	44
30	4	33	20	13	10	40
30	5	33	20	13	20	38
30	6	33	20	13	8	38
30	7	33	20	13	9	38

30	8	33	20	13	9	38
30	9	33	20	13	8	38
30	10	33	20	13	10	38

VI. Conclusion

In this paper, we proposed a mixed integer program to model the eVTOL charging and passenger serving scheduling process. The proposed model is able to provide good computation performance in providing optimal schedule plans for eVTOL real-time operations. Numerical study results also suggest that longer schedule horizon is of great benefit in help reducing average passenger delay without great sacrifice of computation time. In addition, we also find that the effect of increasing number of charging facilities may be limited as of reduce passenger waiting time because those delays may result from insufficient supply of eVTOLs. However, enough number of charging facilities should be equipped to ensure the real-time operation scheduling. These results provide significant insights for both system planner and service providers regarding charging facility capacity design as well as scheduling operations.

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